ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE C · MARKING GUIDELINE

MARKS: 80

Marking guideline for the Academic Central NC(V) Level 4 Electronics and Digital Concepts for Robotics paper 2 · practice c.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Drawings and breadboard layouts may look different and still be correct: check every component, symbol, label and connection. Deduct 1 mark for each wrong connection and 2 marks for any short circuit (no less than 0 for a question).

QUESTION 1

AC supply and transformer T1 ✓✓; bridge rectifier BR1 with AC inputs from the secondary ✓ and + and − outputs correctly connected ✓; C1 across the unregulated supply with correct polarity ✓; 7805 with IN, GND and OUT correctly connected ✓✓; C2 across the 5 V output ✓; R1 and LED1 across the 5 V output with correct LED polarity ✓; all labels and values shown ✓ (10)

[10]

QUESTION 2

Correct circuit: LDR from +9 V to the divider point; potentiometer (wired as a variable resistor) from the divider point to 0 V; 2,2 kΩ resistor from the divider point to the base; emitter to 0 V; buzzer from +9 V to the collector; diode across the buzzer with its cathode to +9 V.

9 V battery symbol with polarity ✓; LDR symbol from + to the divider point ✓; potentiometer as a variable resistor to 0 V ✓✓; 2,2 kΩ base resistor ✓; NPN transistor symbol with collector, base and emitter correct ✓; emitter to 0 V ✓; buzzer between +9 V and the collector ✓; diode across the buzzer, cathode to + ✓; labels and values ✓ (10)

[10]

QUESTION 3

3.1 +9 V and 0 V lines ✓; LDR and R1 divider ✓✓ to the inverting input (pin 2) ✓; VR1 as a potential divider ✓ with its wiper to the non-inverting input (pin 3) ✓✓; op-amp symbol with + and − inputs ✓; pin 8 to +9 V ✓ and pin 4 to 0 V ✓; output (pin 1) to R2 ✓; LED1 to 0 V with correct polarity ✓; labels, pin numbers and values ✓✓; crossing lines drawn without a junction dot ✓ (15)

3.2 Placement ✓✓✓✓✓ (LM358 across the centre gap, LDR, R1, VR1, R2, LED) = 6. Polarity / orientation ✓✓✓: IC notch / pin 1; LED; battery = 3. Connections ✓✓✓✓✓: pin 8 to +9 V and pin 4 to 0 V; LDR–R1 divider to pin 2; VR1 ends to +9 V and 0 V; VR1 wiper to pin 3; pin 1 to R2 and LED; LED to 0 V = 6. (15)

[30]

QUESTION 4

4.1 555 drawn as a block with all eight pins numbered ✓✓; pins 8 and 4 to +9 V ✓; pin 1 to 0 V ✓; R3 pull-up from +9 V to pin 2 ✓ and S1 from pin 2 to 0 V ✓; R1 from +9 V to pins 6 and 7 (joined) ✓✓; C1 from pins 6 / 7 to 0 V with correct polarity ✓; C2 from pin 5 to 0 V ✓; pin 3 through R2 to LED1 ✓ and LED to 0 V ✓; labels and values ✓✓; crossing lines without dots ✓ (15)

4.2 Placement ✓✓✓✓✓✓ (555 across the centre gap, R1, R2, R3, C1, C2, S1, LED – any six correctly placed) = 6. Polarity / orientation ✓✓✓: IC notch / pin 1; C1; LED = 3. Connections ✓✓✓✓✓✓: pins 4 and 8 to +9 V and pin 1 to 0 V; R3 and S1 at pin 2; R1 to pins 6 and 7 (joined); C1 from pins 6 / 7 to 0 V; C2 at pin 5; pin 3 to R2 and the LED = 6. (15)

[30]

TOTAL: 80

END OF MARKING GUIDELINE